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Assignment 03

ITAI-3377 A.I at the Edge

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The Liverpool Smart Pedestrians Experiment

Introduction:

The Liverpool Smart Pedestrians project exudes an effective blend of ingenious technology and progressive city planning. This project tackles the digital evolution occurring in Liverpool, a developing area outside Sydney in New South Wales, Australia. Notably, this effort included university partnership with the University of Wollongong. In an era where cities are quickly adopting new technologies, this undertaking merges data analytics, Internet of Things systems, and environmentally conscious urban design. The initiative extends beyond traditional city monitoring by employing advanced sensors to transform how walking pattern information can support worldly improvements in city planning. Research shows this method conveys a thoughtful approach to balancing surveillance computing with necessary privacy protections; an equilibrium many smart urban projects find challenging.

Objective:

The Liverpool project developed from recognizing that old city planning methods simply can not handle today's complex urban challenges. Growing population density strains current infrastructure, while changing movement patterns require more upscale monitoring than traditional traffic analysis offers. How can cities adapt their structure as population influx continues? How do urban planners balance immediate needs with long term sustainable solutions? "Smart cities" as implemented in this project uses technology not as isolated solutions but as platforms addressing human needs through sustainable tech ecosystems. This people-centered approach makes sure digital tools serve social and economic goals, improving life quality while meeting today's environmental requirements. By combining sensors with existing structures, IoT comes to the rescue! The design showcases a practical tech strategy while acknowledging practical limitations of urban transformation. To build upon current investments rather than replacing them, while creating financially viable solutions that are scalable.

Methodology

The proposal used in this project marks a pivotal shift in urban monitoring systems. My technical evaluation highlights a phased, community focused method that blends public involvement, technological advancement, and iterative design. This structured strategy ensures a well rounded implementation of urban sensing technology, balancing renovation with "on-the-ground" applicability.

Requirements and Constraints: A Human-Centered Technical Approach

The Liverpool project took a different path than typical tech initiatives that start with fixed technical requirements. Instead, it began with thorough community engagement in recognizing the vital connection between technology capabilities and what citizens actually need. Community workshops functioned as collaborative spaces where input from the people directly shaped the engineering design and functions.

This community-enforced method produced requirements balancing technical performance with broader public benefits:

1. *Multi model detection and tracking*: This need for monitoring various travel modes reflects advanced computer vision applications in urban settings. Moving beyond traditional systems that only track vehicles, this versatile approach uses complex neural networks to identify and follow diverse movement patterns. This technology helps city planners spot interaction points between different transportation types, improve signal timing for various travelers, and create architecture that supports multiple transportation needs.
2. *Privacy compliance*: The emphasis on privacy addresses a key challenge in smart city development. By making privacy compliance mandatory for core design, the project team showed their dedication to ethical technology use. A privacy first methodology required elevated edge computing systems capable of extracting valuable insights from visual data while protecting individual identities. This offers an answer to a difficult ethical dilemma.
3. *Leveraging existing infrastructures*: The need to work with current CCTV networks shows practical understanding of technology implementation within city budget constraints. This approach presents a fresh view of technological progress; where new capabilities grow by enhancing existing systems instead of completely replacing them. This upgrade strategy recognizes the major technology investments cities have already made and creates a path to gradually transform these systems into more adaptable sensing networks.
4. *Scalability and interoperability:* The forward thinking requirement for a system that welcomes future tech integration proves a comprehensive grasp of the scientific lifecycle management. By focusing on open architectures and compatible designs, the project

sought to create not just a temporary solution but a lasting platform for continued work. This way it guards against technology becoming outdated and allows the system to evolve with new sensing methods.

Edge-Computing Device: Architectural Revolution

1. Hardware Architecture: Choosing the NVIDIA Jetson TX2 as the main computing platform was a thoughtful selection. This hardware offers an ideal balance of processing power and energy efficiency. Two key factors for edge devices in urban settings with potential power restrictions. The Jetson TX2's specialized neural processors allow complex vision algorithms to run in real time without extreme power use. Adding the Pycom LoPy 4 module shows insight into urban IoT communication needs. LoRaWAN technology provides benefits for city scale systems: excellent range, minimal power needs, and ability to penetrate building materials that block standard wireless signals. This dual communication approach enables flexible deployment across various urban settings; from areas with good wired infrastructure to locations where wireless is the only option.
   The weatherproof IP67 aluminum enclosure tackles the practical challenge of protecting electronics outdoors. Beyond shielding components, the aluminum design acts as a built in cooling system. Very essential for processors handling intensive video analysis.
2. Edge Computing Paradigm: Processing video directly on devices instead of sending raw footage to central servers advances the process of urban sensing. A transformation of standard CCTV takes place where passive recording systems become smart nodes that extract useful details from the data that is collected.This edge outlook does more than save bandwidth (though this helps in urban areas where networks may be limited). When processing data locally and only sharing derived information, the system fundamentally improves its privacy. Visual information never leaves the device and this reduces security risks and privacy concerns.
3. *Processing Pipeline*: The sensor's workflow shows modern integration of computer vision techniques. The 20 frames-per-second rate balances temporal detail with computational efficiency. The sequential process: acquiring frames, detecting objects, tracking them, and managing data; forms a time saving operation where each step builds on the previous one. The tracking algorithm maintains object identity between frames, enabling the system to understand movement patterns rather than just counting objects. This capability transforms static observations into dynamic insights about how different transportation modes interact with urban structures over time.
4. *Data Transmission Strategy:* The adjustable transmission interval reflects careful consideration of trade offs between data timeliness and network resources. Administrators can modify transmission frequency to suit different needs; from traffic management to long term planning. The 5-minute minimum interval for LoRaWAN

transmissions acknowledges the protocol's duty cycle limits; a regulatory necessity for ensuring fair access to shared wireless spectrum.

Validation and Performance:

Evaluation of Validation Experiments The validation tests measured sensor performance using the Oxford Town Center Dataset, featuring high-definition video (1920 × 1080 @ 25 fps) of a busy urban area. The assessment examined accuracy, processing speed, and system resource usage.

Accuracy

- The sensor achieved an average accuracy of 69%, with a median relative error of 33%.
- The system typically undercounted detections, resulting in measurement errors.
- Performance was better when tracking smaller groups but declined in crowded situations due to people blocking each other.
- Detection results maintained a linear relationship with ground truth, confirming the sensor could track trends, though missed detections occurred more frequently than false positives.

Speed (FPS)

- The average processing rate reached 19.57 frames per second, ranging between 4.63 and 22.99.
- Processing speed decreased when more objects appeared due to increased demands on the SORT tracking algorithm.
- The SORT algorithm created a performance bottleneck since it ran on the CPU rather than using GPU acceleration.

System Utilization

- GPU resources were sometimes underused, falling below 40% utilization when the tracking algorithm shifted workload to the CPU.
- CPU usage increased when GPU utilization dropped, highlighting inefficiencies in the current performance.
- Memory and disk usage remained consistent throughout testing.
- Network bandwidth consumption was minimal, as only metadata (counts and movement paths) was transmitted instead of complete video frames.

Conclusion: The validation tests reveal that the sensor delivers moderate accuracy (69%) but tends to undercount detections. The system processes video at an average rate of 19.57 FPS, with performance declining when more pedestrians appear. Future enhancements should prioritize adapting the tracking algorithm for GPU processing to eliminate the CPU blockage. Network efficiency makes the sensor well suited for on the spot execution.

Real-World Applications Analysis

The case study examines two practical implementations of the sensor technology: emergency evacuation monitoring in an indoor setting and traffic monitoring in an outdoor urban environment. These deployments illustrate the technology's potential to enhance urban planning through the monitoring of pedestrian and vehicular movements.

Indoor Deployment: Emergency Evacuation Analysis

- The sensor was installed within the SMART Infrastructure Facility at the University of Wollongong, tracking pedestrian movement on the first floor.
- During the one hour observation period, the system tracked 631 individuals.
- An unplanned fire alarm provided valuable evacuation behavior data:
    - Detection patterns showed two distinct peaks; before and after the evacuation
    - Zero detections during the alarm period, confirming complete evacuation
    - Fewer returning occupants after the all clear, suggesting post alarm behavioral shifts
- Urban Planning Applications:
    - Capability to identify unusual crowd movement patterns
    - Generation of movement visualization tools including heat maps and trajectory data
    - Potential integration with emergency notification systems for immediate alerts

Outdoor Deployment: Liverpool Urban Mobility Monitoring

- A network of 20 sensors was deployed throughout Liverpool's central district, focusing on high traffic pedestrian areas with crosswalks.
- The week-long monitoring period (February 20-27, 2019) captured data on pedestrians, vehicles, and cyclists.
- Key Observations:
    - Consistent daily activity patterns with peak movement between 8:00 and 16:00
    - Vehicle traffic peaked at 9:00 and 16:00, aligning with typical commuting patterns
    - Pedestrian activity showed daily variation with reduced movement on Sunday
    - The system detected 20,399 distinct objects on February 23, 2019
    - Spatial analysis revealed:
        - Pedestrian crossings occurring outside designated crosswalk zones
        - Limited bicycle detection, corresponding with community input on cycling habits

    - Movement patterns for pedestrians and cyclists were mapped to show directional trends
- Urban Planning Applications:
    - Data informed crosswalk placement and pedestrian safety enhancements
    - Traffic flow evaluation and congestion identification
    - Edge computing integration enabling responsive traffic management

Both scenarios show the practical value of sensor monitoring for urban planning. The indoor application proved effective for analyzing emergency response patterns, while the outdoor deployment provided evidence based intel for traffic and pedestrian management. With edge computing integration, municipalities can utilize these analytics to upgrade safety, mobility, and urban design strategies.

Challenges

The Liverpool Smart Pedestrians project encountered several challenges that required thoughtful solutions. Processing pipeline limitations emerged when the SORT tracking algorithm created a computational restraint, reducing system performance during periods of high pedestrian density. Accuracy constraints manifested particularly in crowded urban settings, where the 69% detection rate revealed limitations in handling visual occlusions. The technical team proposed GPU acceleration optimization as a primary solution pathway, acknowledging that shifting tracking operations from CPU to GPU would substantially improve processing. Integration with the existing urban environment presented practical deployment challenges, addressed through the project's dual communication approach that accommodated various installations. Privacy preservation requirements necessitated innovative edge computing architectures that extracted meaningful mobility data while protecting individual identities.

Future Work and Improvements

- Refining Detection Accuracy: Strengthening object recognition methods stands at the forefront of upcoming improvements. Measuring system miscalculations allows for adjustment strategies that correct detection errors, leading to more reliable pedestrian, vehicle, and bicycle counts. Improved calibration techniques further tighten result consistency, ensuring data insights remain dependable for urban mobility decisions.
- Hardware Upgrades for Performance Gains: Current processing operates through NVIDIA Jetson TX2, but shifting to the NVIDIA Xavier platform introduces a leap in capability. The increased CUDA core count significantly amplifies processing efficiency, cutting down delays while boosting overall computational power. This transition lays the groundwork for more seamless urban tracking systems, expanding scalability without overhauling existing arrangements.

- Strategic Software Optimization: Present implementation builds on PyTorch 1.0, chosen for its prototyping flexibility but potentially limiting execution speed. Moving operations toward TensorFlow with TensorRT or Caffe explores alternatives that prioritize computational efficiency over development convenience. Additionally, refining the SORT tracking algorithm to operate within GPU accelerated landscapes align software with evolving hardware capabilities.
- Enhancing Deployment Versatility: Current system design integrates smoothly with existing CCTV networks, but future expansions focus on increasing deployment adaptability. Exploring alternative low-bandwidth transmission methods, such as LoRaWAN, ensures sensor installations remain functional even in areas with limited internet connectivity.

Each step in this improvement path builds on prior investment, shaping a progressive transition rather than an abrupt overhaul. The aim remains clear: getting further in sensor driven urban intelligence without compromising feasibility, affordability, or integration potential.

Evaluation

The Liverpool Smart Pedestrians project stands as a noteworthy advancement in community response, particularly through its exemplary public engagement protocols that established functional priorities aligned with genuine community requirements. Unlike unsuccessful ventures such as the Sidewalk Toronto development, which ultimately dissolved amid confidentiality apprehensions and criticisms of inadequate public consultation, Liverpool's methodology illustrates how digital solutions must be anchored in substantive dialogue with inhabitants to secure legitimacy and adoption. The architectural design successfully navigates the balance between evidence that is backed by urban planning and privacy protection through its pioneering edge computing strategy. Many possibilities exist to improve system precision beyond the current 69% rate through modern computer vision algorithms. While practical applications validate the concept's feasibility, subsequent versions would benefit from greater sensor distribution to deliver more comprehensive spatial coverage and integration with supplementary data streams like atmospheric conditions and transit schedules. The program's most valuable contribution to smart city methodologies lies in showing that technology must be paired with equivalent investment in community engagement and ethical considerations to create truly responsive urban spaces.

References:

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